

Global Facilities - Design & Construction Standard - EUV Basis of Design

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1.0 Purpose

- 1.1 This document serves as a EUV Basis of Design document that provides specifications that shall be followed during the Design, Engineering and Construction of EUV Cleanroom Space, Facilities systems, and EUV Tools as part of new Greenfield FAB Construction projects or retrofit/expansion to existing Yellowfield FAB sites.
- 1.2 This Standard is applicable for all Facilities systems associated with EUV Tools.
- 1.3 The document is intended to supplement the actual Construction Drawings and Construction Specifications for Construction projects.
- 1.4 For each EUV project, the latest EUV information from vendors and suppliers shall be obtained, verified, and used for the project Design, Engineering and Construction.

2.0 Scope

- 2.1 The document covers the new Basebuild System installations and some aspects of EUV tool installation.
- 2.2 This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites for new Greenfield FAB Construction projects or retrofit/expansion to existing Yellowfield FAB sites.

3.0 Definitions and Acronyms

- **AMC** – Air molecular contamination
- **CDA** – Compressed Dry Air
- **CFD** - Computational Fluid Dynamics
- **CR** - Cleanroom
- **EUV** - Extreme ultraviolet lithography
- **FAT** – Factory Acceptance Test
- **FFU – Fan Filter Unit**
- **FMCS/SCADA** - Facility Monitoring and Control System / Supervisory Control and Data Acquisition
- **GHVAC** – General Heating, Ventilation, and Air Conditioning
- **HPCDA/PCDA – High pressure CDA**
- **HVAC** – Heating, Ventilation, and Air Conditioning
- **IST** – Integrated System Test
- **ITP** – Inspection and Test Plan
- **LOTO/COHE** - Lockout Tagout / Control of Hazardous Energy
- **MAU – Make up Air Unit**
- **N+1+F** - System sizing concept for equipment quantity consisting of N duty units, +1 redundant standby unit, +1 future unit
- **OAT** – Operational Acceptance Test
- **PCW – Process Cooling Water**
- **PDP** - Pressure Dew Point
- **PLC** - Programmable Logic Controller
- **POC/LPOC** - Point of Connection / Lateral Point of Connection
- **SAT** – Site Acceptance Test
- **SMS** - Sag Mitigation System
- **T&H – Temperature and Humidity**
- **TUM** - Tool Utility Matrix
- **UPS** - Uninterruptible Power Supply
- **UPW** - Ultra Pure Water
- **VSD/VFD** - Variable Speed Drive / Variable Frequency Drive
- **XCDA** - eXtra Clean Dry Air
- **FCT** - Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKM
- **GFTT** - Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
- **GFC** - Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
- **Global Facilities Central Team** - This is used interchangeably with Innovation & Integration Team
- **Global Facilities Construction Team** - This is used interchangeably with Global Construction Team
- **Global Facilities Operations Support Team** - This is used interchangeably with Global Facilities Operations Team
- **Global Facilities Planning Team** - This is used interchangeably with Global Facilities Line Planning Team

4.0 Roles and Responsibilities

Roles	Responsibilities
GFTT Lead (Discipline Engineer)	Develop and maintain System Standard document Ensure Best Known Methods (BKM), lessons learned, and emerging technologies are documented in System Standard • Perform periodic document review and update System Standard
FCT Lead (Discipline Engineer)	Review and provide technical feedback on System Standard • Support the Site teams by providing knowledge and technical expertise
Site Project Construction Teams – Basebuild and Tool Install	Review and provide technical feedback on System Standard Incorporate System Standard into Project Engineering, Design, Construction, and Tool Install Document project deviations, exceptions, and exclusions from System Standard • Share new lessons learned to GFTT Lead, to determine if it is Best Known Method (BKM) that can be captured in future revision of System Standard
Site Facilities Teams	• Use System Standard as technical reference

5.0 Overview of Systems

- 5.1 This Basis of Design (BOD) document contains the information needed and requirements for designing the EUV Cleanroom Spaces, Facilities Systems serving EUV tools, and some aspects of the EUV tool installation. The EUV tool includes but is not limited to EUV Scanner, EUV Track and H2 abatement unit.
- 5.2 This BOD will specify the following items:
- 5.3 Structural Building Requirements
 - 5.3.1 Facilities Requirements
 - 5.3.1.1 Cleanroom Design
 - 5.3.1.2 Electrical supply
 - 5.3.1.3 Gas supply
 - 5.3.1.4 Water supply and drain
 - 5.3.1.5 Chem. supply and waste collection

- 5.3.1.6 Vacuum supplies and Abatement
- 5.3.1.7 CDA supply
- 5.3.1.8 PCW supply
- 5.3.1.9 Exhaust supply
- 5.3.2 Crane and Tools foundation Requirements
- 5.3.3 Installation Sequence and Coordination
- 5.4 This BOD covers information regarding the pre-installation, installation, service and operation for EUV tools Scanner, Track and H2 abatement.

6.0 Structural Building Requirements

6.1 General building requirements

This section gives the general building requirements.

6.1.1 Building height:

- To support the SEMI standard for factory filling ratio of EUV Platform, it is recommended to use a three-level factory design (consisting of Cleanroom level, Clean Subfab level, and Utility (Dirty) Subfab level) with strict environmental requirements for each level. In this way, all supporting cabinets can fit in the footprint of the EUV Mainframe Exposure Unit (EU).

Table 6.1 Reference Building height for ASML EXE 5000

		Height [m]
Cleanroom	with Automated Reticle Transport (ART)	Contact ART supplier and Crane supplier ^[a]
	without ART	Contact Crane supplier ^[a]
Subfab		≤ 10.86 ^[b]
	free height below obstructions ^[c]	≥ 2.3 ^[d]
Utility Fab		≤ 18.5 ^[e]
	free height below obstructions ^[c]	≥ 2.3 ^[d]

^[a] More information can be found in the TWINSKAN EXE:5000 Crane Requirements Manual (CRM).

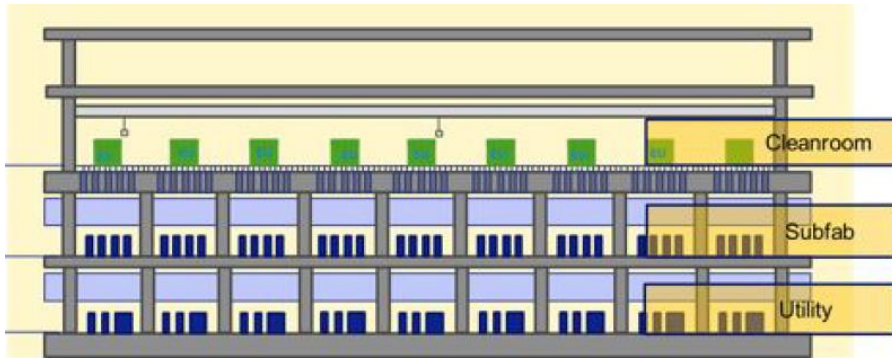
^[b] See Z7 in Figure 2.2

^[c] See Figure 2.1.

^[d] Excluding the pre-vacuum-abatement system. Contact the pre-vacuum-abatement system supplier for more details.

^[e] See Z8 in Figure 2.2.

Figure 6.1 illustration of Building levels



- For Cleanroom level height, need to consider Tools, AMHS, Crane requirement to decide the height from Raised floor to ceiling.
- For Plenum height, need to consider space required for return air, people maintenance.
- For Raised floor height, need to consider Tools interconnection, and hook up requirement.
- For Clean Subfab level height, need to consider Tools height, interconnection, and hook up requirement. The Subfab height shall be minimum 6.55m clear height (in example, 7.25 slab to slab height with 0.7m FAB level waffle slab thickness).
- For Utility Subfab (Dirty Subfab) height, need to consider abatement height, interconnection, and hook up requirement. The Utility Subfab height shall be minimum 5.1m clear height (in example, 5.7m high slab to slab with 0.6m Subfab level waffle slab thickness)

6.1.2 Cleanroom Ceiling Height

- The Cleanroom Ceiling Height shall be minimum 8.5m high clear, to accommodate EUV Scanner, Cleanroom Crane, and ART automated delivery system.
- OHT vehicle hoist belt max limit is 5.5m. It is recommended to check EUV reticle load port height for CR ceiling height over 8.5m.

6.1.3 Raised Floor Height

- The Raised Floor Height shall be minimum 800mm high.

Table 6.2 Reference Project V and F11E building Heights

	EUV Fab height comparison		unit: mm
Location	F11E	F16 V	Recommendation
Utility Subfab (Dirty)	5500	4900	5500
Subfab (Clean)	7250	7700	7300
Raised floor height	800	800	800
Cleanroom Ceiling	8500**	8000	8500**
Plenum (Truss)	6000	3000	Depend on design

*All heights stated are "Slab to Slab" heights, and include the floor slab thickness and structural beams/waffle

**F11E Cleanroom ceiling height has the capability to raise to a maximum 10,000mm height in future. It also recommended for new EUV fab.

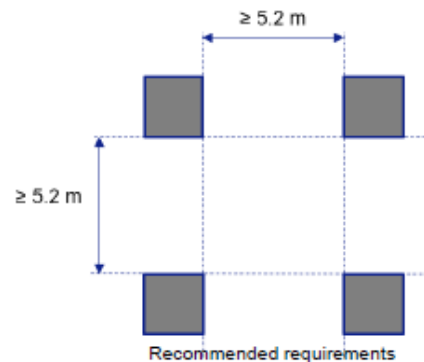
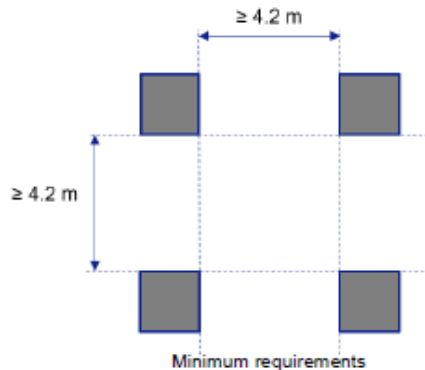
6.1.4 Waffle slab structure – column spacing and slab openings

- Subfab level Column spacing (column to column) shall be 5.2m clearance, to allow sufficient space for installation and maintenance of Tool cabinets and equipment.
- Utility Subfab level Column spacing (column to column) shall be 5.2m clearance, to allow sufficient space for installation and maintenance of Abatement Unit, Tool cabinets, and equipment.

Table 6.3 Reference Pillar grid requirements for ASML Scanner EXE 5000

	Pillar [m]
Minimum ^[a]	$\geq 4.2 \times \geq 4.2$
Recommended	$\geq 5.2 \times \geq 5.2$

[a] The Drive Laser position is inflexible and asymmetric within the pillar grid.



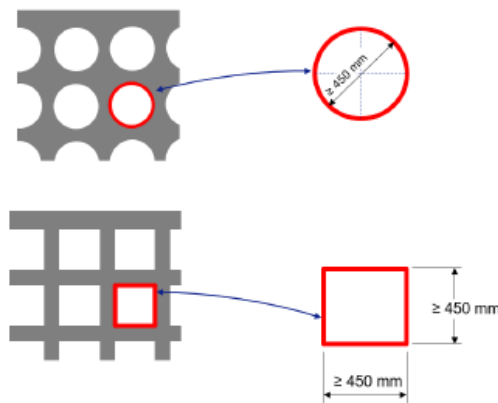
- Waffle slab openings at FAB level floor slab (between Subfab level and FAB level) shall be 450mm diameter large, to allow passthrough of tool hookup utilities routing.
- During construction, waffle slab openings shall be temporarily covered with removable plates, for fall protection.

	Feed-through hole [mm]
Minimum Subfab to Cleanroom	≥ 310
Recommended ^[a] Subfab to Cleanroom Utility Fab to Subfab	≥ 450 ^{[b] [c]}

[a] For pre-vacuum facility piping (DN320) including mounting tolerances and excluding factory construction tolerances.

[b] For 350 mm < D < 400 mm, there is a risk of welding during pre-installation.

[c] For 310 mm < D < 350 mm, typical 2 x DN250 expected locally for pre-vacuum piping, instead of 1 x DN320.



6.1.5 Stiffness limits

A building suitable for the EUV installation must comply with the stiffness limit requirements. For the detail information, please check latest EUV IRM.

6.1.6 Vibration Limits

- FAB level vibration design shall meet VC-D⁺ requirement, 200μin-sec.
- Clean Subfab level vibration design shall meet VC-C requirements, 500μin-sec.
- Dirty Subfab level vibration design shall meet VC-C requirements, 500μin-sec, due to some EUV cabinet will locate in this area.
- Vibration specification graphs are given for the Exposure Unit support frames and the equipment subfab floor. If the measured results are below the lines indicated in these graphs, the vibrations are within specification and will not influence the system performance. If, at some frequencies, the measured results are above the lines, calculations must be done to determine the effect on the system.
- Below is reference Vibration limits for example ASML EXE:5000 EUV Scanner tool.

Figure 6.2 Vibration limits – Cleanroom

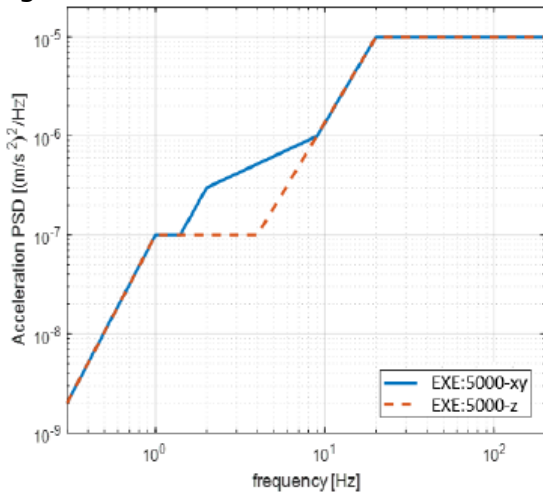
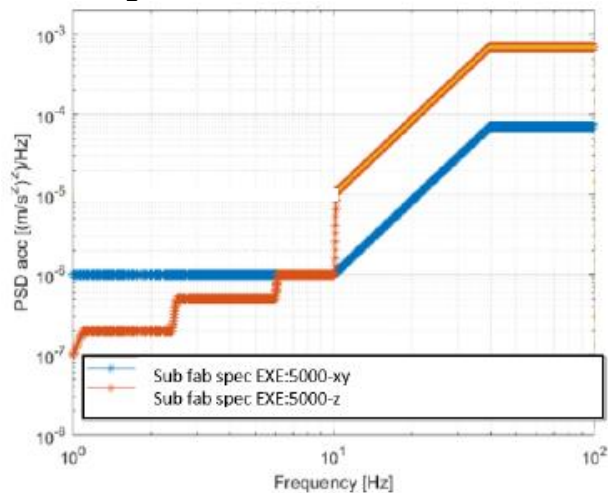


Figure 6.3 Vibration limits – Subfab



6.1.7 Equipment loading for Building structural design

1. Building Structure Floor loading requirement for EUV in main Fab is 32kN/m².
2. Crane loading and area, the loading for HiNA crane is 47,350 kg/set
3. Cleanroom Ceiling for AMHS loading and area, the latest information is 600 kg/m².
4. Building Structure Floor loading for EUV in Subfab is 1,500 Kg/m²
5. Building Structure Floor loading for abatement in dirty subfab is 1,500 Kg/m²
6. CR Raised floor loading request by EUV should be higher than 1500Kg/in². It recommends install 3000Kg/in² for blank panel and 2400kg/in² for perforated panel.

6.2 Floor Flatness Requirements

- 6.2.1 For RC floor in Cleanroom, subfab and dirty subfab, the flatness should < 3mm within 3m x 3m area.
- 6.2.2 For RC floor in Cleanroom, subfab and dirty subfab, the flatness should < 3mm within 3m x 3m area.
- 6.2.3 The table below is the requirement from EUV manual for reference.

Table 2.10 Floor requirements - Loading dock to Cleanroom including install/service area

Flatness [mm]		Slope [mm/m]		Load ^[a] [kg]			Raised Access Floor (RAF)	
Move-in path ^[b]	Service area EU	Move-in path		Point load [kg/sq. inch]	Point load [kg/ø50 mm]	Distributed load [kg/ø533 mm]	Height ^[c] [mm]	Tile height [mm]
		/1 m	/10 m					
NEN 2747 Class 4	< 2	< 4	< 10	≥ 1500	≥ 2575 ≥ 3500 (WS SiSo lanes only)	≥ 1850	450 - 1500	≤ 50

^[a] For the part of the move-in path that is built as a Raised Access Floor (RAF): per tile (600 mm x 600 mm) at any position on the tile.

^[b] Required over the shadow footprint of the move-in part (~ 7.5 m x 2.9 m).

^[c] Height is defined as the distance from the Cleanroom side of the waffle slab to the top side of the Raised Access Floor (RAF) tile.

Note

The Raised Access Floor (RAF) should only be supported on the four corners of tiles in the pre-install routing area and the Beam Transport System (BTS) install/service area of the Exposure Unit (EU).

Table 2.11 Floor requirements - Loading dock to Subfab/Utility Fab including install area

Flatness [mm]	Slope [mm/m]		Load [kg]	
Move-in path ^[a]	Move-in path		Point load ^[b]	Distributed load
	/ 1 m	/ 10 m		
< 4	< 4	< 10	≥ 1700	t.b.d.

^[a] Required over the shadow footprint of the move-in part.

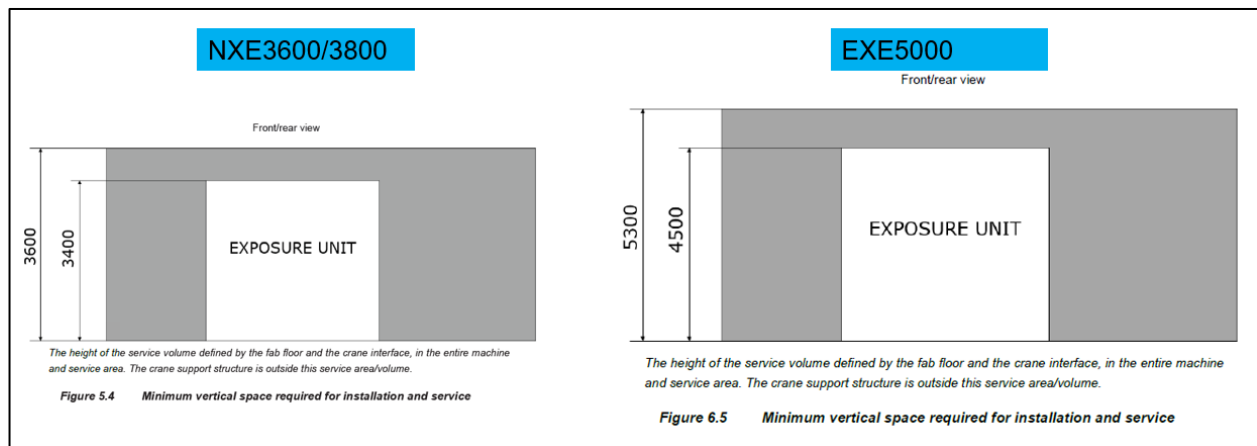
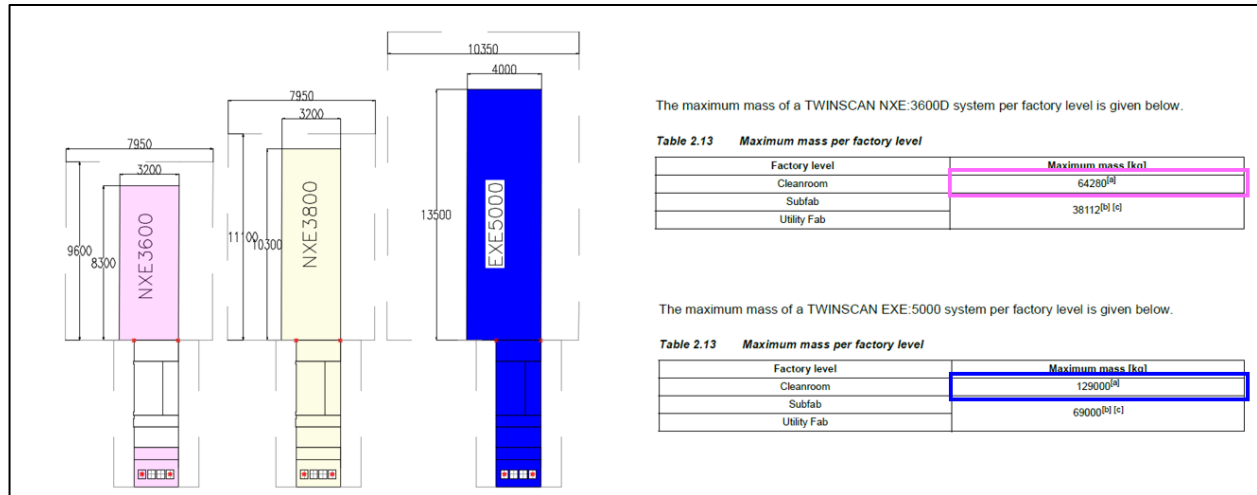
^[b] At 645 mm² (square 25.4 mm / square 1") at any position on the tile .

6.3 Dimension Requirements

This section gives the EUV Scanner and Track tool dimensional requirements.

6.3.1 EUV Scanner Dimension Requirements

- ASML NXE:3600 – Tool Dimension 8.3m Length x 3.2m Width x 3.4m Height
- ASML NXE:3800 – Tool Dimension 10.3m Length x 3.2m Width x 3.4m Height
- ASML EXE:5000 – Tool Dimension 13.5m Length x 4.5m Width x 4.5m Height



6.3.2 EUV Track Dimension Requirements

- TEL Lithius Pro Z Track Model 2G 1.0BL – Tool Dimension 7.890m Length x 2.485m Wide x 3.500m Height
- TEL Lithius Pro Z Track Model 3G 1.0BL – Tool Dimension 7.890m Length x 2.485m Wide x 3.500m Height
- TEL Lithius Pro Z Track Model 2G 1.5BL – Tool Dimension 8.595m Length x 2.485m Wide x 3.500m Height
- TEL Lithius Pro Z Track Model 2G 1.5BL – Tool Dimension 10.190m Length x 2.485m Wide x 3.500m Height

TEL ProZ Track Model	2G	3G	2G 1.5 Block	2G 1.5 Block
Capable WPH	165	140	240	240
Dimension (Length in mm)	7890	7890	8595	10190
NXE3600 Compatible (160wph)	Yes	Yes	Yes	Yes
NXE3800 Compatible (240wph)	No	No	Yes	Yes

7.0 Facility requirements

7.1 Cleanroom Design

Table 7.1: FAB EUV Cleanroom Specification – Temperature & Humidity

Parameter	Specification Requirement	Applicable Spaces
EUV Cleanroom Particles	ISO 14644-1 Class 5	FAB EUV Cleanroom Area
EUV Reticle Cleanroom Particles	Minimum: ISO 14644-1 Class 3 Preferred: ISO 14644-1 Class 2	FAB EUV Reticle Area
EUV Cleanroom Temperature	22.5 ± 1 °C [72.5 ± 1.8 °F]	FAB EUV Cleanroom Area and EUV Reticle Area
EUV Cleanroom Humidity	Design Capability: 42 ± 3 %RH DRAM Spec: 43 ± 3 %RH	FAB EUV Cleanroom Area and EUV Reticle Area
EUV Cleanroom Pressurization	Minimum: 9 Pa Recommended: 13-15 Pa	FAB EUV Cleanroom Area and EUV Reticle Area
EUV Clean Subfab Temperature and Humidity	20~25 °C [72.5 ± 1.8 °F] RH: 30~55 %	Due to PCW supply temperature of EUV tools is 15.5°C, the environment dew point needs to be controlled under 15.5°C.
EUV Utility (Dirty) Subfab – for abatement and/or EUV subtools	18~25 °C [72.5 ± 1.8 °F] RH: 30~55 %	Due to PCW supply temperature of EUV tools and abatement is 15.5°C, the environment dew point needs to be controlled under 15.5°C.
EUV Facility Utility (Dirty) Subfab Particles	ISO 14644-1 Class 9	FAB EUV facility subfab Area

Table 7.2: FAB EUV Area and Reticle Area Cleanroom Design

Space	ISO Class	Heat Load	Minimum FFU Coverage	Minimum Air Change Rate
FAB EUV Cleanroom Area	ISO Class 5	Recommended 900 W/m2	Minimum: >50% Recommended: >50%	Minimum: > 90 ACH Recommended: > 90 ACH
FAB EUV Reticle Area	ISO Class 3	Recommended 450-750 W/m2	Minimum: 100% Recommended: 100%	Minimum: > 150 ACH Recommended: > 150 ACH

7.1.1 EUV Cleanroom Design Heat Load shall be 900W/m2 default value.

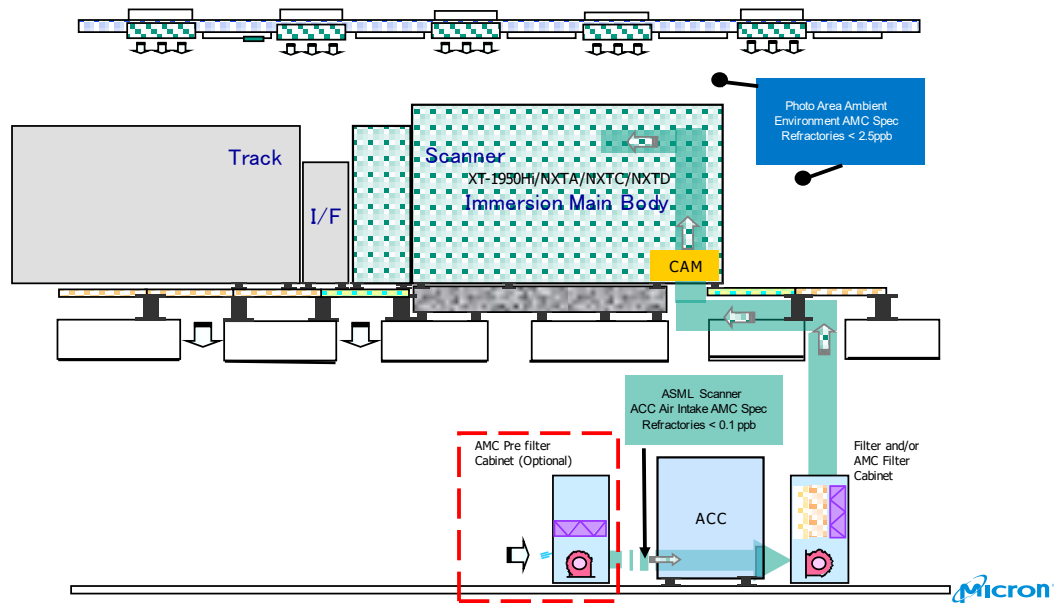
7.1.2 EUV Cleanroom FFU coverage shall be minimum 50%.

7.1.3 AMC Filters shall be installed for every FFU in the EUV Cleanroom Area and EUV Reticle Area. These AMC Filters shall be installed after the completion of the initial Cleanroom construction, and before the first Photo Scanner tool is moved into the Cleanroom Space.

7.1.4 Cleanroom AMC requirement and monitoring method please refer to the table below. For latest information, please refer to the Global design & construction standard. Global Facilities - Design & Construction Standard - [\(General - Critical Specification Alignment for Technology Node Transfer \(micron.com\)\)](#)

Parameter	Specification Requirement	Remarks
Acids		Photo Area – Acids, based on speciation
- Fluoride	< 5 ppbv	
- Chloride	< 2 ppbv	
- Nitrite	< 10 ppbv	
- Nitrate	< 1 ppbv	
- Sulfate	< 1 ppbv	
Critical Specs for ASML Warranty – AMCs		
Acids, Total	< 5 ppbv	ASML ACC Air Intake
Ammonia/Amines, Total	< 25 ppbv	
Condensables	< 50 ppbv	
Refractories	< 0.1 ppbv	

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7.1.5 EUV AMC requirements please refer to table below.

Parameter	Cleanroom	ACC air intake	Subfab ^[a]	Conditioned Utility ^[a]	Utility (facility) ^[a]
Temperature range (min - max) [°C]	20 - 24	20 - 25	20 - 26	18 - 30	5 - 40
Temperature variation ^[b] [°C/h]		< 1		< 5	≤ 10
Thermal load to environment [kW]	≤ 16	N/A	≤ 25	≤ 40	
Altitude [m]	0 - 2000				
Air pressure [kPa(a)]	75 - 105 ^[c]				
Pressure range [kPa]	nominal value ± 3			nominal value ± 10	
Pressure variation ^[b]	[kPa/h]	≤ 2		≤ 15	
	[Pa/10 s]	≤ 10	N/A		
Relative humidity range [%]	30 - 60 ^[d]				20 - 90 ^[d]
Relative humidity variation ^[b] [%/h]	≤ 10	non-cyclic changes: ≤ 0.5%/5 min		≤ 10	N/A
		cyclic changes: see Figure 5.2			
Cleanliness [ISO class]	6 or better	7 or better		9 or better	N/A
Allowable concentrations of contaminants ^[e] [ppbv]:					
• Total volatile acids		≤ 5	N/A	N/A	N/A
• SO ₂ in total volatile acids		≤ 0.3			
• Total volatile base		≤ 50			
• NH ₃ in total volatile base		≤ 25			
• Condensable organics		≤ 50			
• Non-condensable organics		≤ 1000			
• Refractory compounds		≤ 0.1			

[a] The Utility Data Sheet (UDS) specifies the possible installation locations of the cabinets for Utility (with mandatory CSRs), Conditioned Utility, and Subfab.

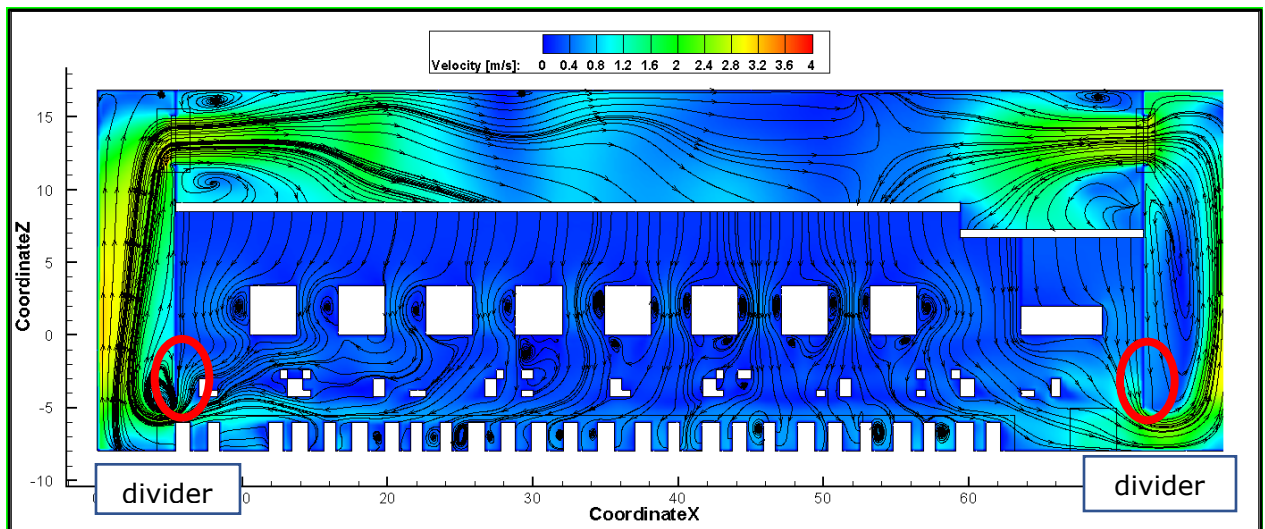
[b] Variation is defined as the difference between max. and min. values in the given time window.

[c] The maximum air pressure difference between the cleanroom (EU) and the equipment subfab (HPDL) must be <10 Pa.

[d] Under no circumstances condensation is allowed. For example, there must be no condensation when the subfab or (conditioned) utility air comes into contact with the PCW supply and return pipes.

[e] These specifications conform with the charcoal filter warranty requirements to give a filter lifetime of at least 6 months.

- 7.1.6 During Design phase, the Cleanroom conditions shall be checked using Computational Fluid Dynamics (CFD) modelling software. CFD modelling shall be used to simulate and determine the following Cleanroom conditions, and make the necessary improvements to the Design.
- 7.1.7 It is recommended to extend the EUV Scanner ACC Cabinet Clean Air Intake via ducted intake, up to a higher level directly below the L30 Cleanroom Floor slab or directly below L30 Cleanroom Ceiling FFU discharge, to provide more stable temperature, humidity, and filtered air conditions entering the ACC Cabinet. The air intake location should be 5m away from Develop cabinet if it locates in CR level. If the air intake is below cleanroom floor slab, it is recommended to install air temperature and humidity sensor at air intake to monitor air status.
- 7.1.8 It is recommended to design and install dedicated supplemental GHVAC cooling in the Clean Subfab level with ducted supply distribution to the temperature critical cabinets, due to high heat load and to provide stable temperature conditions. These cabinets include the ASML Scanner AERC / ALSE cabinets.
- 7.1.9 It is not required to install environmental ionizer in EUV Cleanroom area.
- 7.1.10 Cleanroom Ceiling design should consider crane lifting point. The required loading and installation of lifting point need to refer to the manufacturer's latest information.
- 7.1.11 It is recommended to design return air shaft divider partition in Clean Subfab level at the upper portion of the Return Air Chase opening, to draw the return air from lower section of the Return Air Chase and have better air flow pattern if dry cooling coil installed in the plenum.



7.2 Electrical Supply:

The mains supply system shall operate within the limits of the supply voltage range in accordance with the NFPA 79, IEC 60204-1, SEMI S22, and SEMI F47 standards.

- 7.2.1 EUV Scanner Main Body and Components listed (PA0C, PA1C, PA2C, PA3C, TRIC, AC Power Box, System THC) Power Supply shall be connected to Sag Mitigation System (i.e. SMS, DVR). SMS protection shall provide Voltage rebuild in 40-60% for 1~2 seconds. (Depends on the power sag historical at site).
- 7.2.2 EUV Scanner MDRC, PDU_MK2, HPDL-MC, and BC20M Component Power Supply shall be UPS.
- 7.2.3 EUV Abatement Unit and Components Power Supply shall be connected to Sag Mitigation System (i.e. SMS, DVR)
- 7.2.4 EUV Track Main Body and Components Power Supply shall be connected to Sag Mitigation System (i.e. SMS, DVR).
- 7.2.5 Main supply requirements:

	EU (Scanner) ^[a]	EU (Source) ^[a]	HPDL-MC	High Power Drive Laser (HPDL)
Configuration	3-phase 4-wire Y-configuration L1 + L2 + L3 + N + PE TN-S with solidly grounded neutral conductor			3-phase 3-wire Y-configuration L1 + L2 + L3 + PE TN-S solidly grounded
Operating voltage [VAC] ^[b] Nominal	LN = 230 LL = 400			LL = 400 OR LL = 480
Operating voltage range [VAC]	400 ± 10%			400 ± 10% OR 460 ± 10%
Nominal operating frequency [Hz]	50 or 60 (configured at customer site)			
Operating frequency range [Hz]	± 2%			
Phase order ^[c]	L1/L2/L3 clockwise			
Voltage harmonic distortion (THDu) (IEC 61000-2-4)	< 10% - class 3 - industrial grids			
Current harmonic distortion (THDi)	< 30% at full load conditions			
Phase unbalance (NFPA 79/IEC 60204)	< 2%			
IEC Installation Category (IEC 60664/UL 3111)	category IEC class II or T2			
Fast transients/burst voltages immunity (IEC 61000)	± 2 kV			
Surge voltages immunity (IEC 61000)	± 1 kV L-L ± 2 kV L-PE			

	EU (Scanner) ^[a]	EU (Source) ^[a]	HPDL-MC	High Power Drive Laser (HPDL)
Voltage sags immunity	according to SEMI F47			
Earthing continuity (NFPA 79/IEC 60204/SEMI S22) Continuity of the protective bonding circuit test (NFPA 79/ IEC 60204/SEMI S22)	0.1 Ω			
Full load current per connection [A]	50 - 500			
Full load power [kVA]	625	100	60	1400

[a] Mains supply for Scanner and Source need to be provided by the same grid or transformer.

[b] For the supply cable and connections to the system, comply with local safety requirements or standards.

[c] If multiple machines use the same power grid, the load can be balanced more evenly by using different phase order connections for adjacent machines. The clockwise phase order must be maintained. Examples of phase order connections: machine 1 - L1/L2/L3; machine 2 - L2/L3/L1; machine 3 - L3/L1/L2; machine 4 - L1/L2/L3 etc.

It is recommended to take diversity factor as table below for Electrical capacity sizing.

NXT3600				
Function	Voltage	Rate KVA	Diversity	Demand KVA
Scanner	400	735	0.6	441
Laser	480	1,400	0.6	840
Track	208	226	0.4	90.4
Edward	480	233	0.4	93.2
AMHS	208	278	0.4	111.2

7.2.6 Arc flash safety requirements

For EUV tools, the item below needs to fulfill the following requirement

- The arc flash incident energy shall be/remain below 1.2 cal/cm² so that customer or ASML service engineers do not need to wear arc flash Personal Protective Equipment (PPE).

The active/valid NFPA-70E/IEEE1584 must be used for the calculation of the arc flash incident energy, considering the parameters shown in Table 6.1.

- The active/valid NFPA-70E/IEEE1584 must be used for the calculation of the arc flash incident energy, considering the parameters shown as below.

ASML parameters for arc flash calculations

Parameter	Value
Conductor gap	25 mm (1 inch)
Working distance	455 mm (18 inches)
Equipment type	MCCs and panels
Arc type	Arc in a box
Grounding type	Grounded system

- Calculations must be performed by the customer for all fab mains supply connections: HPDLMC, TRUC, PAXC, MDRC, BC20M and PDU.

ASML parameters for arc flash calculations	
Parameter	Value
Conductor gap	25 mm (1 inch)
Working distance	455 mm (18 inches)
Equipment type	MCCs and panels
Arc type	Arc in a box
Grounding type	Grounded system

- The tripping time of MCCB to be considered in 0.02 second for the feeder of distribution panel supply to EUV panels.
- Calculations must be performed for all mains supply connections: HPDLMC, TRUC, PAxC, BC20M, AMDC 1, AMDC 2, PDU, and PMDC.

7.2.7 Electromagnetism

The TWINSKAN NXE:3600D system shall comply with the following standards regarding Electro-Magnetic Compatibility (EMC):

- Emission Standards:
 - EN 55011:2009 Class A, Group 1 >20 kVA
 - FCC CFR 47 Part 18:2018
- Immunity Standards:
 - IEC61000-6-2:2005
- Electrical equipment for measurement control and laboratory use- EMC requirements:
 - EN 61326-1:2013

7.2.8 Power supply recommendation:

Tool	Component	Power supply	Phase_Type	Frequency	Voltage_Type
EUV	MDRC	UPS	R.S.T.N	50/60	400V-230V
EUV	PDU_MK2	UPS	R.S.T.N	50/60	400V-230V
EUV	BC20M	UPS	L-N	50/60	100-230V
EUV	HPDLMC	UPS	R.S.T.N	50/60	400V-230V
EUV	PA0C	SMS	R.S.T.N	50/60	400 or 480V
EUV	PA1C	SMS	R.S.T	50/60	400 or 480V
EUV	PA2C	SMS	R.S.T	50/60	400 or 480V
EUV	PA3C	SMS	R.S.T	50/60	400 or 480V
EUV	TRUC	SMS	R.S.T	50/60	400 or 480V
EUV	ELRC	SMS	R.S.T.N	50/60	400V-230V
EUV	MID	SMS	R.S.T	50/60	400V
EUV	MID	SMS	R.S.T	50/60	400V
EUV	EU	SMS	L-N	50/60	120V
EUV	EU	SMS	L-N	50/60	120V
Abatement	EUV ZENITH Gen3 (SRC)	SMS	R.S.T	50/60	480V
Abatement	EUV ZENITH Gen3 (EXP)	SMS	R.S.T	50/60	480V
Abatement		SMS	L-N	50/60	120V
Abatement		SMS	L-N	50/60	120V
Track	AC power box	SMS	R.S.T	50/60	208V
Track	System THC	SMS	R.S.T	50/60	208V
Track	RCM LOC	SMS	L-N	50/60	120V

Track	AC power box	SMS	R.S.T	50/60	208V
Track	System THC	SMS	R.S.T	50/60	208V
Track	Fan	SMS	L-N	50/60	120V
Track	Fan	SMS	L-N	50/60	120V
Track	Fan	SMS	L-N	50/60	120V

7.2.9 Power supply segregation: Power Supply distribution to multiple EUV tools shall be segregated across minimum 2 or more Distribution Transformers, to prevent the failure of 1 Transformer to cause shutdown of all EUV tools.

7.3 Gas supplies

7.3.1 General

- For the EUV model(s) to be installed, refer to the most recent manufacturer's "Facility and Infrastructure Requirements" manual for the EUV model for the latest utility gas requirements and purity specifications.
- Existing gas capacities shall be reviewed to determine if there is sufficient supply for the EUV, or if expansion is required.
- Review quality of existing gas supplies to determine if additional purification is required to meet the EUV specifications.
- Gas supply systems shall be designed according to the local codes and regulations, fire protection, supplier, and site risk assessment site and corporate EHSS guidelines.
- Refer to Global Facilities – Gas – UHP Stainless Steel for Bulk and Specialty Gases Design Standard for pipe/tube materials of construction.
- Refer to Global Facilities – Gas – Bulk Gas Supply and Distribution System (N2, O2, Ar, He, H2, CO2) Specifications for system sizing, design and configuration.
- Refer to Global Facilities – Gas – Cylinder Specialty Gas and Bulk Specialty Gas Supply Systems for specialty gas system design and configuration.
- Refer to Global EHS – Toxic Gas Monitoring and Double Containment Standard for gas detection requirements and alarm setpoints.

ASML FIR Requirements Example:

Table 3.8 H₂ requirements	
	Requirements
Total nominal flow [Nl/min] ^[a]	660
Pressure range [kPag]	400 - 999
Temperature range [°C]	cleanroom ± 5
Cleanliness grade ^[b] [°]	9 N or better
Additional ASML requirements [ppbv]:	
C _x H _y (45 - 100 amu)	< 0.2
C _x H _y (> 100 amu)	< 0.002
Refractory Compounds	< 0.001
O ₂	< 0.25
H ₂ O	< 1
CO	< 1
CO ₂	< 0.25
Σ all others (incl. C _x H _y < 45 amu)	< 1
Cleanliness particles (ISO 14644-1:2015)	class 3 or better

[a] Consumption at facility supply GFU [Nl/min].

[b] The specified H₂ cleanliness purity can only be obtained with a purifier installed in the H₂ supply line as close as possible to the GFU. A purifier can be obtained from e.g. Entegris. With the proper purifier installed, H₂ purity graded 5.5 N can be applied as input to the purifier (SEMI C58-1116).

[c] 2000 ppbv is the N₂ content in H₂ per definition of 5.5 N (SEMI C58-1116). No N₂ removal is required.

7.3.2 Hydrogen (H₂) Supply, Distribution, Purification, and Recovery**Hydrogen Supply System**

- Basebuild Hydrogen Capacity shall be carefully studied to ensure that there is sufficient Hydrogen supply to the EUV tools, due to the high loading demand.
 - Hydrogen systems shall be designed, fabricated and tested in accordance with recognized national pressure vessel and piping codes (i.e., ASME B31.12, Hydrogen Piping and Pipelines).
- Electrical equipment shall be installed in accordance with NFPA 70.
- Electrical equipment and the piping system shall be bonded and grounded to dissipate static electricity. All grounding apparatus shall comply with NFPA 70, National Electrical Code®. Lightning protection in accordance with NFPA 780 should be considered.
- Hydrogen supply systems shall include Emergency Isolation Valves (EIVs). EIV is typically located at the source supply container/tank/tube trailer to permit emergency isolation of the system. EIVs shall be capable of full closure to ensure no gas leakage. EIVs may be operated remotely or locally (at the valve) and shall have fail-closed operation. EIVs are to be automatically closed for excess flow, hydrogen gas leak detection or UVIR at source system, and earthquake detection.
- Hydrogen supply systems shall include excess flow valves/switches that shut off flow when the built-in sensing mechanism detects that a present minimum flow rate has been reached. The EFV/EFS is used to provide an immediate hydrogen shutoff in the event of failure of the piping system or equipment.
- Hydrogen gas detectors shall be installed to alarm and activate shutdown if gas leakage is detected.

- For purity concerns, hydrogen pipelines are to be fully welded construction with the exception of VCR connections contained within exhausted enclosures monitored for gas leaks.
- Because of the leak potential, all underground hydrogen piping shall be welded construction with no valves, mechanical joints, or connections installed underground. Underground piping shall be buried to a depth in accordance with the local codes for protection from frost, surface loads, and shifting due to unstable soil. A protective casing shall be provided for pipe routed under railroad tracks and may be required by local codes under roadways or wherever the piping can be subject to physical damage. The casing shall be open at both ends or adequately vented. The soil shall be tested for corrosiveness to determine the necessity for cathodic protection and/or a corrosion protective coating. Underground piping having cathodic protection should be connected to aboveground piping through an electrically insulated joint to isolate the cathodic protection system.
- It is recommended to purchase the Hydrogen Recovery System with the Edwards' Abatement unit for EUV to reduce the overall H2 consumption.

Hydrogen Distribution System

- Hydrogen is an HPM gas and shall be distributed within the fab using Distribution Valve Boxes (DVBs) and Valve Manifold Boxes (VMBs).
- DVBs/VMBs enclosures are to be exhausted and monitored for gas leaks.
- Refer to Global EHS – Toxic Gas Monitoring and Double Containment Standard for alarm setpoints and shutdown configuration.

Hydrogen Purification

- Consult with Entegris for the appropriate model of regenerable H2 purifier necessary given the inlet quality to meet the manufacturer's required H2 specification to the tool.
- H2 purifiers shall be located in a purifier room that complies with the relevant code requirements for flammable storage (including fire protection, room UVIR monitoring, damage limiting construction, fire walls, etc.)
- Buildings and doorways shall be of fire resistant construction as determined by national codes or regulations.
- Explosion relief blowout panels shall be provided in exterior walls or roofs and should be designed so that if an explosion occurs the pressure will be relieved safely.

Hydrogen Recovery

- It is recommended to purchase the Hydrogen Recovery System with the Edwards' Abatement unit for EUV to reduce the overall H2 consumption.
- If required by the module, a point-of-use hydrogen purifier (getter type) may be installed between the Edwards' blender box and the EUV for moisture removal.
- Hydrogen Demand without Edwards HRS:

$$\text{Total H2 Demand} = (\text{Source}_{\text{H2}} + \text{Scanner}_{\text{H2}}) * N * 1.2$$
- Hydrogen Demand with Edwards HRS:

$$\text{Total H2 Demand} = \{[(0.2 * \text{Source}_{\text{H2}} + \text{Scanner}_{\text{H2}}) * (N-1)] + [\text{Source}_{\text{H2}} + \text{Scanner}_{\text{H2}}]\} * 1.2, \text{ where}$$

N = number of EUVs to be installed,
 $\text{Source}_{\text{H2}}$ = H2 flow rate for EUV Source chamber,
 $\text{Scanner}_{\text{H2}}$ = H2 flow rate for EUV Scanner chamber,
1.2 = multiplier factor to provide for additional 20% buffer capacity.

Assumptions: HRS is capable of recovering 80% of the source H2 flow rate, and one HRS is inoperable.

Hydrogen CQC

- Install a CQC system to analyze the purified hydrogen for O2, moisture, CO, CO2, and THC to the ppb levels as identified in the respective Facilities and Infrastructure Requirements manual for the model EUV(s) to be installed.

7.3.3 Nitrogen Supply and Distribution

- Existing N2 supply shall be reviewed to determine if there is sufficient supply for the EUV, or if expansion is required.
- Review quality of existing N2 to determine if existing purifiers can meet the EUV specifications, and is able to support the additional capacity. Determine if additional purifiers are required for N+1 reliability.
- If new, separate and dedicated N2 purifiers are required for the EUV, install a dedicated N2 CQC to analyze and monitor for O2, moisture, CO, CO2, and THC to the ppb levels as identified in the Facilities and Infrastructure Requirements manual for the model EUV(s) to be installed.

7.3.4 Argon, Helium, and Carbon Dioxide Supply and Distribution

- Review existing gas supply capacities to determine if there is sufficient supply for the EUV, or if expansion is required. Flow rates are small such that existing supply systems may be able to support the EUVs.
- Ensure quality of existing gas supplies can meet the requirements as identified in the Facilities and Infrastructure Requirements manual for the model EUVs to be installed. Additional purifiers of

the same type as used by existing facilities may be required for N+1 reliability.

- New, dedicated purifiers may be required to meet the EUV requirements. For new, dedicated purifiers, also install CQC to analyze and monitor O₂, moisture, CO, CO₂ (except for CO₂ gas), and THC impurities to the ppb levels as identified in the Facilities and Infrastructure Requirements manual for the model EUV(s) to be installed.

7.3.5 2%Hydrogen/Argon Mix Supply and Distribution

- 2%H₂/Ar is a non-HPM gas and shall be supplied from cylinder racks.
- The cylinder racks shall be N+1 configuration for reliability.
- The 2%H₂/Ar distribution system shall be by main and laterals with lateral point-of-connection (POC) valves.
- An end of line pressure transducer to be installed and report back to Micron's FMCS.
- CQC is not required for 2%H₂/Ar. Cylinder Certificate of Analysis (CoA) to be provided by gas supplier.

7.4 Process cooling water (PCW)

7.4.1 PCW specification shall reference the Global PCW Design and Construction Standard: [\(Global Facilities - Design & Construction Standard - Mechanical - Process Cooling Water System \(micron.com\)\)](#)

7.4.2 Required flow and pressure shall be regulated to meet tool's demand

7.4.3 PCW Supply Temperature shall be 15.5°C.

7.4.4 The Clean Subfab and Utility (Dirty) Subfab environmental conditions (temperature and humidity) and GHVAC systems shall be designed to meet the Specs provided in above sections, and to prevent condensation of the PCW systems.

7.5 Ultra-Pure Water (UPW)

7.5.1 UP W specification will be reference to Global UPW Design and Construction Standard: [\(Global Facilities - Design & Construction Standard - UPW & WWT - DIW System.pdf\)](#)

7.5.2 Required flow and pressure shall be regulated to meet tool's demand.

UPW supply requirements

	Requirements
Pressure range [kPag]	200 - 500
Supply temperature range [°C]	19 - 27
Suspended particle size [µm]	≤ 30
Calcium hardness [ppm]	≤ 100
pH balance	6 - 8
Electrical conductivity [µS/cm]	≤ 1

UPW treatment requirements

Supporting cabinet	HXAL	HXCU
Manufacturer	NALCO	
Product	Water Care - Aluminium (anti corrosion)	Water Care - Copper (anti corrosion)
Concentration	50 ± 50% ml/100 l	5 ± 50% ml/100 l

7.5.3 All TWINSCAN EXE:5000 drains are gravity style drains used for non-routine service actions or in case of a leakage. Some drains might be temporarily pressurized by CDA during service actions in order to remove all the water in the applicable circuit.

7.6 Chemical supply for Track (done by Sankar)

7.6.1 The chemical types and consumption used for each EUV tool shall be determined and verified with Photo Process Area and IE/Layout Teams.

7.6.2 For reference, below is an example of number of chemicals and its classification. Actual chemical that will be used is confidential, hence, detail information will be provided during project execution. The consumption & POC information provided is for reference only as it is subject to change depending on Photo Track tool model. All information shall be validated with Micron for accuracy and/or changes prior to final design.

Global Facilities - Design & Construction Standard - EUV Basis of Design

Last Modified: 08/17/22

Version: 1

ID: A3YRXSD74VDV-57553043-527

S/N	Chemical Category	Process	Bulk Chemical	POC per Track	Consumption (LPM)		Design & Material Requirements				
					Peak Flow	Ave Flow	Chemical System Type	Tank Material (Min Req)	Valve / Wetted Parts	Piping Material	Filter Membrane
1a	Caustic	CAR	Chem A	NA	NA	NA	CCB & BCTU	SS / PTFE-ET (2.4mm)	Teflon	Primary - PFA 451 Secondary - Clear PVC	Entegris, PTFE 50nm NXCZM1C0R0D (10") NXCZM2C0R0D (20")
1b	Caustic	CAR	Chem B	NA	NA	NA	CBU	SS / PTFE-ET (2.4mm)	Teflon	Primary - PFA 451 Secondary - Clear PVC	NA
1c	Caustic	CAR	Chem B	2	3	0.14	CDM & VMB	SS / PTFE-ET (2.4mm)	Teflon	Primary - PFA 451 Secondary - Clear PVC	PALL XpressKeen G2 2nm ABFG1XP23EH11(10") ABFG2XP23EH11(20")
1d	Caustic	CAR	Chem C	NA	NA	NA	CTU	NA	Teflon	Primary - PFA 451 Secondary - Clear PVC	NA
2a	Solvent	CAR / MOR	Chem D	NA	NA	NA	CCB, BCTU	SS316L EP	SS316L EP	SS316L EP	Entegris Torrento X 2nm TRCJE110R0D (w/o key/10") TRCJE210R0D (w/o key/20")
2b	Solvent	CAR / MOR	Chem D	4	3	0.02	CDU & VMB	SS316L EP	SS316L EP	SS316L EP	Entegris Torrento X 2nm TRCJE110R0D (w/o key/10") TRCJE210R0D (w/o key/20")
3	Solvent	CAR	Chem E	1	3	0.02	CTDM & VMB	SS316L EP	SS316L EP	SS316L EP	Entegris Torrento X 2nm TRCJE110R0D (w/o key/10") TRCJE210R0D (w/o key/20")
4	Acid	CAR	Chem F	1	3	0.08	CTDM & VMB	SS / PTFE-ET (2.4mm)	Teflon	Primary - PFA 451 Secondary - Clear PVC	PALL XpressKeen G2 2nm ABFG1XP23EH11(10") ABFG2XP23EH11(20")
5	Solvent	CAR	Chem G	1	1	0.01	CTDM & VMB	NA	SS316L-EP	SS316L-EP	Entegris, PTFE 50nm NXCZM1C0R0D (w/o Key / 10") NXCZM2C0R0D (w/o key/ 20")
6	Acid	CAR	Chem H	1	3	0.08	CTDM & VMB	SS / PTFE-ET (2.4mm)	Teflon	Primary - PFA 451 Secondary - Clear PVC	PALL XpressKeen G2 2nm ABFG1XP23EH11(10") ABFG2XP23EH11(20")
7	Solvent	CAR	Chem I	1	0.16	0.0012	CTDM & VMB	NA	SS316L EP	SS316L EP	Entegris, 15nm NXCWG1C0R0D (10") NXCWG2C0R0D (20")
8	Corrosive Solvent	MOR (Inpria)	Chem J	1	3	0.05	CTDM & VMB	SS / PTFE-ET (2.4mm)	Teflon	Primary - Anti-static PFA Tubing Secondary - SS304	Toronto X 2nm TRCJE110RKD (w key/10") TRCJE210RKD (w key/20")
9	Solvent	MOR (Inpria)	Chem K	1	3	0.03	CTDM & VMB	SS316L EP	SS316L EP	SS316L EP	Toronto X 2nm TRCJE110R0D (w/o key/10") TRCJE210R0D (w/o key/20")
10	Corrosive Solvent	MOR (Inpria)	Chem L	1	3	0.03	CTDM & VMB	SS / PTFE-ET (2.4mm)	Teflon	Primary - Anti-static PFA Tubing Secondary - SS304	Toronto X 2nm TRCJE110RKD (w key/10") TRCJE210RKD (w key/20")
11	Corrosive Solvent	MOR (Inpria)	Chem M	1	3	0.03	CTDM & VMB	SS / PTFE-ET (2.4mm)	Teflon	Primary - Anti-static PFA Tubing Secondary - SS304	Toronto X 2nm TRCJE110RKD (w key/10") TRCJE210RKD (w key/20")

7.7 Chemical supply for Reticle Mask Clean SUSS tool

7.7.1 The chemical types and consumption used for each EUV tool shall be determined and verified with Photo Process Area and IE/Layout Teams.

7.7.2 For reference, below is an example of chemicals used for EUV Reticle Mask Clean SUSS tool (Sussmicrotec Masktrack X):

S/N	Chemical Classification	Bulk Chemical	Remark
1	Caustic	DEV 4262 (TMAH 2.38%)	Confirmed requirement
2	Oxidizer	H2O2	Confirmed requirement
3	Caustic	NH4OH	To be determined based on evaluation
4	Acid	H2SO4	Confirmed requirement

7.8 Chemical waste collection system

7.8.1 The Chemical waste types and consumption from each EUV tool shall be determined and verified with Photo Process Area and IE/Layout Teams, based on the actual Chemical type used. Site will need to carry out its own compatibility test of the drain mixture to ascertain compatibility with existing drain streams and/or separate waste collection/treatment systems as per local regulations.

7.9 Vacuum

7.9.1 Process Vacuum (PV)

PV specification will be reference to Global PV standards. Required flow and pressure shall be regulated to meet tool's demand.

7.10 Abatement (H2 removal)

- Utility specification will be reference to Global facility standards. Required flow and pressure shall be regulated to meet tool's demand.
- The PCW pressure requirement of abatement is pressure difference rather than supply pressure. The PCW booster pump is not necessary for the Abatement unit and shall not be provided.

7.11 Exhaust

- 7.11.1 Exhaust specification will be reference to Global Exhaust Design and Construction Standard: (Global Facilities - Mechanical – Process Exhaust System Standard.pdf (micron.com) Required flow and pressure shall be regulated to meet tool's demand.
- 7.11.2 The Basebuild exhaust system shall have sufficient capacity and exhaust pressure at the distribution lateral end-of-line to meet the EUV tool exhaust flow and pressure requirements. Exhaust booster fan(s) shall NOT be installed for the EUV tools. The booster fans may cause potential AMC issues, disruption or fluctuations to the exhaust lateral, result in another potential point of failure, additional maintenance requirements, and additional space requirements.

7.12 Compressed Air (CDA, HPCDA, and XCDA)

- 7.12.1 CDA specification will be reference to Global CDA Design and Construction Standard: (Global Facilities - Design & Construction Standard - Mechanical - Compressed Air System Standard (micron.com) Required flow and pressure shall be regulated to meet tool's demand.
- 7.12.2 Compressed Air supply for EUV Scanner shall be supplied from Photo CDA / High Pressure CDA (PCDA/HPCDA) system.
- 7.12.3 Compressed Air supply for EUV Track shall be supplied from CDA system.
- 7.12.4 eXtra Clean Dry Air supply for EUV Scanner shall be supplied from Purified CDA (XCDA) system. The XCDA spec. should Refer to the manufacturer's "Facility and Infrastructure Requirements" manual for the EUV model for the latest requirements and specifications.

7.13 Tool Installation

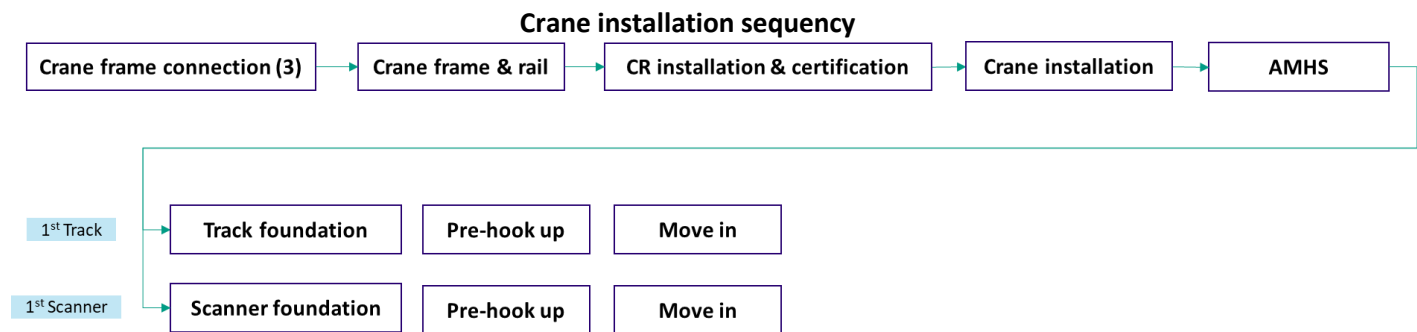
- 7.13.1 Exhaust connections for tool hookup shall use rigid exhaust ductwork to as close as possible to the Tool connection, and then a short section of flexible exhaust ductwork for the final connection. The flexible ductwork length shall be kept as short as possible, with minimal bends and no sagging/low points to prevent liquid accumulation and leakage from the flexible ductwork.

8.0 EUV Installation Sequence and Coordination

- 8.1 The following is a reference EUV Installation sequence and guidelines that can be followed, for EUV construction project:

- 1) Truss connection to Crane frame. (estimated 8 weeks)
- 2) Crane frame and rail installation (estimated 8 weeks)
- 3) Completion of Cleanroom Construction and successful Cleanroom Certification

- 4) Overhead Crane move-in and Installation overhead (estimated 8 weeks)
- 5) AMHS and Reticle ART Installation overhead (estimated 3 weeks)
- 6) EUV Track pedestal base foundation installation (estimated 3 days)
 - a. Begin EUV Track tool pre-hookup installation (estimated 4 weeks)
 - b. Move-in EUV Track
 - c. EUV Track tool final-hookup installation
- 7) EUV Scanner pedestal base foundation installation (estimated 4 weeks)
 - a. Begin EUV Scanner tool pre-hookup installation (estimated 10 weeks)
 - b. Move-in EUV Scanner
 - c. EUV Scanner tool final-hookup installation



9.0 Document Control

9.1 Approval: GLOBAL FAC GFTT APPR

9.2 Notification:

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
GLOBAL_FAC_NOTIFY
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY

9.3 Final Viewer Access

All Micron Team Members at all Sites

9.4 Retention

Adherence to the requirements specified in
[Global Facilities - Document Control Procedures](#)

9.5 Review

Biennially (two years)

2.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New Document	08/17/2022

Appendix 1: Template Form for Site Specific Deviation List**Site:**

No	Specification Section and Item	Deviation Proposal	Scope Impact (Benefit, Cost, Schedule)	Remark	Approval: 1. Site 2. GFTT